

## Problem

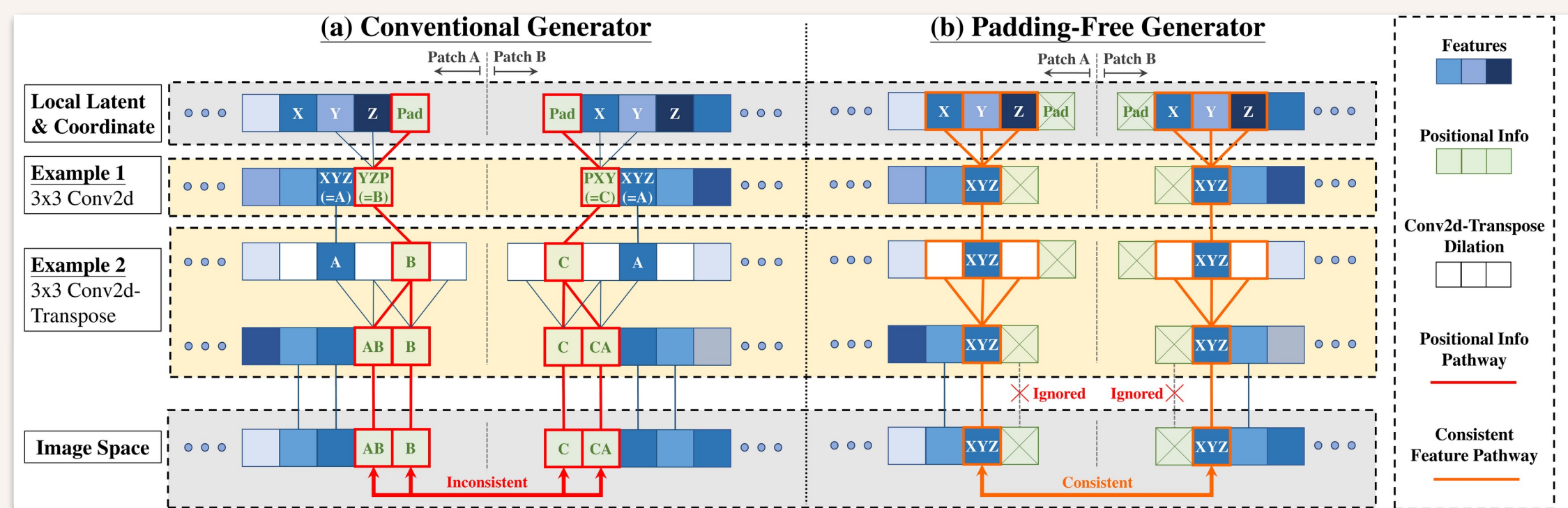
Generating arbitrarily large — even infinite-pixel — images is bottlenecked because existing high-resolution GANs tie **computation, memory, and training-data field-of-view** directly to the output resolution.

*Compute, memory, and field-of-view all scale with resolution — so generators stay locked to the size they trained at.*

## Motivation

- Large images must stay **locally and globally consistent**, avoid repetitive patterns, and look realistic.
- Yet prior generators lean on **zero-padding** for positional cues that break down once the output size differs from training.

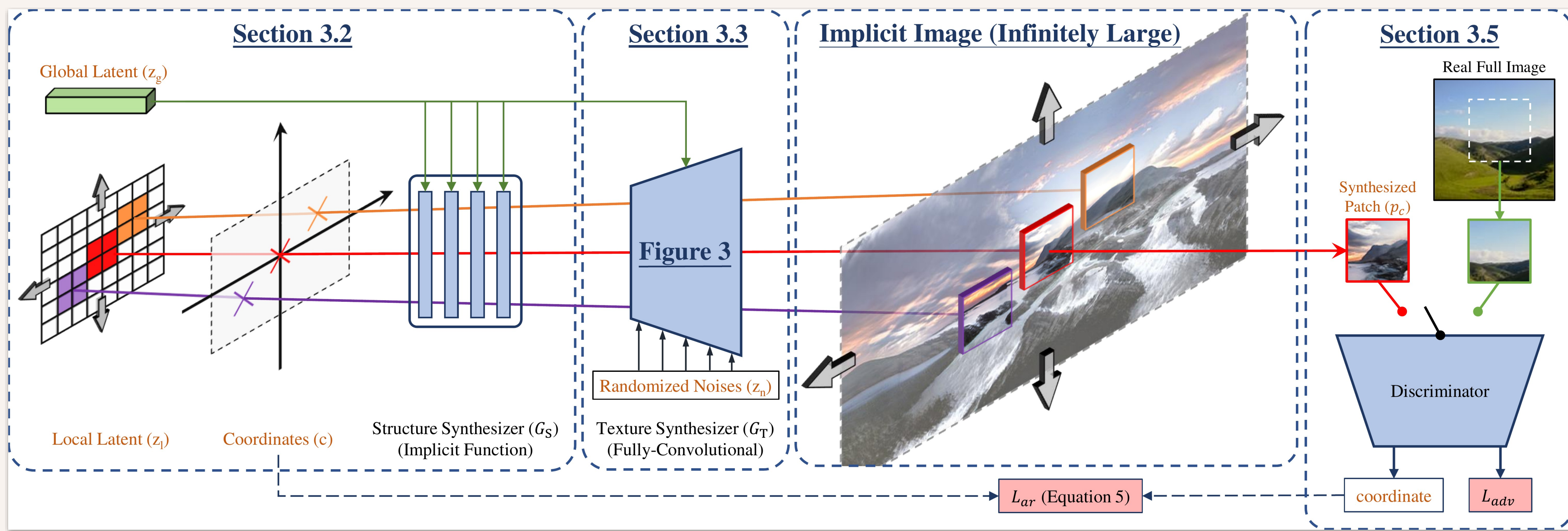
When the border is infinitely far away, padding-derived position cues cannot generalize — image centers fill with drastically repetitive, structurally broken texture.



Conventional generators (left) leak inconsistent pixels from zero-padding; the padding-free generator (right) stays consistent at any position.

## Method

- Structure synthesizer**  $G_S$ , a coordinate-driven neural implicit function, computes  $z_S = G_S(z_g, z_l, c)$  from global, local, and coordinate inputs.
- Padding-free texture synthesizer** renders each patch  $p_c = G_T(z_S, z_g, z_n)$  with all position encoding removed — so patches at the same coordinate are **identical** and tile seamlessly at **O(1) memory**.



Overview: structure synthesizer  $G_S$  and padding-free texture synthesizer  $G_T$  synthesize arbitrary-size images from latents, coordinates, and noise.

## Dataset / Benchmark

- Flickr-Landscape** — a new dataset of **450,000** high-quality landscape images — evaluates arbitrary-size synthesis.
- Outpainting uses **Places365** (62.5K) and **Flickr-Scenery** (54.7K); all models train on 101×101 patches cropped from 197×197 real images.
- Training and inference at **any size** run on a single GTX TITAN X GPU at **constant memory**.

Places365 • 62.5K

Flickr-Scenery • 54.7K

train: 101×101 patches

## Key Results

| SCALEINV FID ↓            | 4×           | 8×            |
|---------------------------|--------------|---------------|
| SinGAN                    | 145.12       | 210.22        |
| StyleGAN2+NCI             | 79.83        | 189.65        |
| <b>InfinityGAN (Ours)</b> | <b>61.41</b> | <b>121.18</b> |

–18.4 ScaleInv FID vs. best baseline at 4×

SO WHAT → Beyond 4× extension InfinityGAN beats the strongest baseline while holding constant O(1) memory as the output size grows, and over 90% of a user study preferred its results.

## Headline Numbers

**61.41**

ScaleInv FID • 4× extension

vs 79.83 best baseline

>90%

human preference

7.20×

faster inference

O(1)

memory • any size

## Ablation Study

- Strip **all padding** from StyleGAN2+NCI and structure collapses: ScaleInv FID 79.83 → **211.22** at 4×
- Restore position through the implicit  $G_S$  and quality recovers: 211.22 → **61.41**.

## Takeaway

UNLOCKS

The same design enables spatial style fusion, multi-modal outpainting, and arbitrary-length panorama & inbetweening.

Disentangling global appearance, structure, and texture — and dropping all padding — lets InfinityGAN tile independent yet consistent patches into images of any size.

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Paper

Code

